Multilingual Sentiment Analysis Using Cross-lingual Embeddings

Authors: Dr. Maria Garcia, Prof. Ahmed Hassan, Dr. Yuki Tanaka
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ABSTRACT:
We propose a novel approach for multilingual sentiment analysis using cross-lingual word embeddings and transfer learning. Our method achieves state-of-the-art results on 15 languages with minimal labeled data per language.

1. INTRODUCTION

Sentiment analysis in low-resource languages remains challenging due to limited annotated data and linguistic diversity.

2. METHODOLOGY

Our approach consists of three main components:

2.1 Cross-lingual Embedding Alignment

We use MUSE (Multilingual Unsupervised and Supervised Embeddings) to align word embeddings across languages in a shared vector space.

2.2 Transfer Learning Framework

- Pre-train on English sentiment data (1M samples)
- Fine-tune on target language (1K samples)
- Use adversarial training for domain adaptation

2.3 Ensemble Methods

Combine predictions from multiple models using weighted voting.

3. EXPERIMENTAL RESULTS

Languages tested: Spanish, French, German, Arabic, Chinese, Japanese, Hindi, Russian, Portuguese, Italian, Dutch, Swedish, Polish, Turkish, Korean

Average F1 scores:
- Our method: 0.847
- mBERT baseline: 0.823
- XLM-R baseline: 0.831

4. ERROR ANALYSIS

Common errors include cultural context misunderstanding and handling of code-switching in social media text.

5. FUTURE WORK

Integration with large multilingual models like PaLM and GPT-4.